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Direct Comparison of (Anthropometric) Methods for the Assessment of Body Composition

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Keywords
Body composition · Body fat · Anthropometry · Dual-energy X-ray absorptiometry · Bioelectrical impedance · Hydrodensitometry · Underwater weighing

Abstract
Objective: Several methods for the assessment of body composition exist, yet they yield different results. The present study aimed to assess the extent of these differences on a sample of young, healthy subjects. We hypothesised that differences in body composition results obtained with different methods will vary to the extent that a subject can be misclassified into different nutritional categories. ***Research Methods and Procedures:*** Underwater weighing (UWW), bioelectrical impedance analysis (BIA), anthropometry (ANT), and dual-energy X-ray absorptiometry (DXA) were used to assess body composition. An extensive list of ANT regression equations (or sets of equations) was analysed in terms of accuracy and precision relative to DXA. ***Results:*** When DXA-determined body fat (BF) values were taken as a reference, UWW overestimated BF in both genders. In contrast, BIA (measured with a given bioimpedance analyser) underestimated BF in females, although BIA-determined BF did not differ from DXA in males. A huge difference in BF estimates (8–29% for females and 6–29% for males, for DXA-determined BF of 25.5% and 13.9% for females in males, respectively) was observed across a number of ANT regression equations; yet, ANT proved not to be inferior to DXA, provided that regression equations with the highest combinations of accuracy and precision were chosen. ***Conclusions:*** The study proved grounds for comparison of body composition results of young, healthy subjects, obtained with different methods and across a wide range of ANT regression equations. It also revealed a list of the most appropriate ANT regression equations for the selected sample and reported their accuracy and precision.
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Introduction

Body composition assessment is crucial for the determination of nutritional status in both healthy individuals and patients. It is also indispensable for progress monitoring in weight loss, fitness assessment, rehabilitation programs and so on [1].

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Table 1. Basic characteristics of the subjects

	N	Age, years	Body height, cm	Body mass, kg	BMI, kg/m^2	Urine-specific gravity
Females	26	21.0 (1.0)	164.7 (5.5)	56.6 (8.7)	20.9 (3.1)	1.012 (0.007)
Males	22	21.2 (1.5)	180.7 (6.0)	74.7 (8.9)	22.8 (1.8)	1.008 (0.007)
p value	–	0.66	0.000	0.000	0.01	0.04

p values denote inter-gender differences (Student *t* test); mean (SD).

Several methods for body composition assessment are available, each with its own advantages and disadvantages (for reviews, see Ackland et al. [2], Duren et al. [3], Wells and Fewtrell [4]). When it comes to laboratory measurements, underwater weighing (hydrodensitometry; UWW) has long been the gold standard for body composition assessment [5]. As UWW cannot be performed in all subjects (e.g., in those, who are not accustomed to water), dual-energy X-ray absorptiometry (DXA) is nowadays often the method of choice for body composition assessment, as it is, for example, able to detect muscular cachexia in patients; of course, being a two-dimensional method, it has its own drawbacks [6]. Also, magnetic resonance (MR) [7], ultrasound measurements [8], computerised tomography (CT) [9], isotope dilution [10], and air displacement plethysmography [11] have all been used to assess body composition in laboratory settings. These, however, are not methods that can usually be performed in the field. Not surprisingly, new field methods, such as bioelectrical impedance analysis (BIA), have gained popularity [12], although the validity of BIA measurements has been questioned [13–15], and thorough scientific evaluation of the method is difficult to perform as different BIA analysers are being used and their underlying equations are inaccessible [16].

Anthropometry (ANT), although having its own disadvantages [1], has been recommended as a field method of choice [17]. Provided the measurements are performed by skilled personnel and according to standardised procedures [18, 19], the accuracy of ANT is in line with that of other, much more sophisticated methods [20]. However, as skinfolds acquired with ANT provide only localised information about the thickness of subcutaneous fat tissue, different regression equations were developed to calculate body fat mass and/or body fat percentage; one would expect they provide rather uniform values, which, however, is not the case [21]. Consequently, any comparison of body composition results obtained with different methods in different studies becomes questionable.

Surprisingly, several different methods for body composition assessment have rather rarely been simultaneously applied to the same group of subjects [22, 23], although this would enable a less biased and more generalised comparison of body composition results between the studies. Also, apart from the study of Robič et al. [24] on pregnant women, data on simultaneous comparison of body composition results obtained with numerous anthropometric regression equations relative to a reference method are scarce; although more than a hundred regression equations exist [1], to our knowledge, a maximum of 11 have been simultaneously compared with a reference method of choice (hydrodensitometry), and even then the comparison was performed for a specific population (professional soccer players) only [21].

The aim of our study was therefore to simultaneously compare body composition results obtained with UWW, BIA, and ANT, with DXA – as reference method. We hypothesised that body composition results obtained with different methods will vary to the extent that the same subject can be misclassified into different nutritional categories.

Material and Methods

The protocol of the study was approved by the Ethics Committee of the Republic of Slovenia (No. 69/03/15), and the study was performed in accordance with the Declaration of Helsinki.

Young (19–25 years old), healthy volunteers were included in the study (Table 1), and they provided informed consent prior to participation. No participating female was pregnant. Their body composition was assessed with 4 different methods, including ANT, UWW, BIA, and DXA, which was adopted as a reference method. A detailed description of individual body composition methods is provided below.

Anthropometry

ANT was performed by an experienced anthropometrist according to standardised procedures [18]. Body mass and body height were measured with a certified medical level scale with an anthropometer (Seca 799, Germany), sitting height with a certified sitting height table (Harpenden, UK). Whenever weighing was

Table 2. Subjects' skinfold thickness

	Skinfold, mm							
	biceps	triceps	sub-scapular	iliac crest	supra-spinal	abdominal	thigh	calf
Females	6.5 (2.3)	14.8 (5.0)	11.0 (4.3)	20.0 (5.8)	9.0 (5.1)	18.5 (6.4)	26.1 (8.9)	14.9 (6.1)
Males	4.2 (0.9)	8.9 (2.6)	9.7 (2.1)	16.6 (5.3)	6.3 (2.1)	14.7 (5.0)	12.3 (6.4)	7.2 (2.1)
p value	0.000	0.000	0.231	0.040	0.025	0.030	0.000	0.000

p values denote inter-gender differences (Student *t* test); mean (SD).

Table 3. Percentage of body fat (mean (SD)) as determined by different methods for the assessment of body composition

	N	Dual-energy X-ray absorptiometry	Anthropometry	Underwater weighing	Bioelectrical impedance
Females	26	25.5 (5.1)	25.4 (2.6)[a]	27.5 (6.2)[x]	22.7 (5.8)
Males	22	13.9 (3.0)	13.8 (2.0)[b] 13.3 (2.3)[c]	15.5 (4.6)[x]	13.6 (3.0)

For anthropometry, the results of methods with the highest combination of accuracy and precision are presented. Females, set of equations of [a]Lean et al. [25] 1 (for body density) and Brožek et al. [32] (for body fat percentage). Males, direct equation of [b]Al-Gindan et al. [30] 2; set of equations of [c]Lean et al. [25] 4 (for body density) and Brožek et al. [32] (for body fat percentage). [x] Two females and 1 male were excluded from underwater weighing (UWW) analysis due to obviously erroneous readings (they were unable to reliably follow the UWW protocol).

performed, mass of previously weighted dry subject's clothes (swimming costume or shorts) was subtracted from measured mass of the subject.

Four breadths (elbow, hip, knee, ankle), 6 girths (upper arm, wrist, waist, hip [gluteal], thigh, calf), and 8 skinfolds (biceps, triceps, subscapular, iliac crest, supraspinal, abdominal, thigh, and calf) were measured. Breadths were measured with a large spreading calliper with rounded ends (DKSH GPM, Switzerland), girths with a non-stretchable narrow measuring tape (Rollfix, Germany), and skinfolds with a calibrated skinfold calliper (Harpenden, UK). Each skinfold measurement was performed 3 times, and median value was adopted as representative (see also Robič et al. [24]).

Numerous regression equations for the calculation of either body density (BD; g/mL) (termed "indirect") or body fat percentage (BF; %; termed "direct") were collected from the scientific literature. Only the published equations suitable for our sample were used and compared. When the same literature source reported more than 1 regression equation, they were numbered consecutively (e.g., Lean et al. [25] 1, 2); the exact equations used are presented in online supplement 1 (for all online suppl. material, see www.karger.com/doi/10.1159/000508514). To avoid inaccurate citations, only the original published literature sources of the regression equations were considered.

The methodology of each literature source was carefully examined and, if needed, appropriate standardisation of regression equations was made. For example, some equivocal terminology was detected for skinfold measurement sites; for example, the term "suprailiac" can be interpreted as referring to both, iliac crest or supraspinal location; therefore, the exact description of the measurement site (or the original reference of the measurement protocol) was looked for and appropriately considered. Also, as different units were used in different regression equations (e.g., body height was referred to in centimetres [26] or in metres [27]) and different units were used to describe the amount of body fat (it was, e.g., expressed as body fat mass [in kilograms] [28], but also as total adipose tissue volume [in litres] [29]), the appropriate conversion factors were used for standardisation in such cases. Following the approach of Al-Gindan et al. [30], a factor of 0.92 was used to convert from adipose tissue volume (L) to body fat mass (kg).

BF was calculated from both direct and indirect equations, the latter in combination with both the equation of Siri [31] and Brožek et al. [32]. A total of 78 different equations, including 22 direct equations and 56 equation combinations (thus, an indirect equation for the calculation of BD, combined with the equation of either Siri [31] or Brožek et al. [32]), were used for BF calculation in females (online suppl. 1a, 2a). A total of 67 different equations, including 23 direct equations and 44 equation combinations, were used to calculate BF in males (online suppl. 1b, 2b). Body mass index (kg/m^2) was also calculated (Table 2).

Underwater Weighing

Prior to UWW, subjects' lung residual volume of air (RV) was determined with whole body plethysmography (MasterScope Body, CareFusion, Germany) by experienced medical personnel.

First, subjects performed slow and forced spirometry to determine their vital capacity (VC). After at least 3 satisfactory manoeuvres, the highest VC value was selected as representative. Subjects then entered a body plethysmograph, which was daily calibrated with a 3-L syringe and for which time constant to detect a leak was measured each morning. Functional residual capacity (FRC) was determined by an occlusion method during normal breathing. Residual volume was calculated using the formula: RV = FRC – ERV, where ERV is expiratory reserve volume measured during spirometry. At least 3 repeated measurements of FRC were performed in the same subject, and their mean was used as representative, provided that the 3 measurements did not differ for >5%.

For UWW, the dry mass of the swimming suit was noted. Subjects were then asked to void themselves, to remove all other clothing, and to don a swimming suit. UWW was performed with a certified hanging scale (Salter Brecknell 235 10S, UK) in a heated pool. Water temperature was measured with a thermometer (Amarell GmbH, Germany) and used for the calculation of water density with an online calculator (http://antoine.frostburg.edu/chem/senese/javascript/water-density.html).

During UWW, subjects sat on a submerged chair attached to a hanging scale, so that their whole body but their head was submerged under water. After maximal expiration, they immersed their head under water and the reading on a hanging scale was noted. The procedure was repeated 10 times for each subject and the median value was taken as representative. Finally, Archimedes' principle and Siri equation [31] were applied to calculate BD and BF, respectively.

Bioelectrical Impedance Analysis

Subjects were instructed to avoid alcohol, heavy exercise, and sauna for 24 h prior to BIA. Just prior to measurement, subjects voided themselves and their urine specific gravity was determined with refractometer (Atago Master-Sur/Nα, Japan) to control for hydration. In case subjects' urine specific gravity exceeded the reference normohydration value of 1.020 [33], they were asked to drink a large glass of water and to wait for approximately 20 min. Then, the whole procedure was repeated and, if the subjects' urine specific gravity was within the reference values for normohydration, BIA was performed with a multifrequency segmental bioelectrical analyser (Tanita MC-780MA, Japan). This analyser is a 3-frequency (5/50/250 kHz) tetrapolar (hands-to-feet) device that enables body composition measurements in the upright posture.

Dual-Energy X-Ray Absorptiometry

DXA was performed in the morning, immediately after BIA measurements, by experienced medical personnel, with a Hologic dual-energy X-ray absorptiometer in the fan beam mode (QDR-4500A Hologic, USA) using the software provided by the manufacturer (QDR for WindowsXP). Full body scan was performed with subjects lying supine for approximately 15 min.

Statistical Analysis

Skinfold thickness and basic characteristics of the subjects were compared between males and females with the Student *t* test for independent samples.

For each method for the assessment of body composition (DXA, BIA, UWW, and ANT [several equations]), average SD BF was calculated, separately for males and females. As 2 females and 1 male were unable to reliably follow the UWW protocol, their UWW results were excluded from further UWW analysis.

For each subject, the difference between BF as assessed with a given anthropometric equation (or equation combination) and that assessed with DXA was determined. Then, the mean difference between DXA-determined BF and each anthropometric method was calculated (separately for males and females). Finally, the absolute values of these differences were first sorted according to size from smallest to largest and then ranked. The anthropometric method with the lowest rank (i.e., with the smallest [absolute] average difference to DXA-determined BF) was considered the most accurate. ANT-determined BF values were compared with the reference DXA-determined BF with a one-way ANOVA for correlated samples.

Standard error of estimate (SEE), a measure that determines how variable the observations are around the reference regression line, was calculated for each anthropometric equation (or set of equations) relative to DXA-determined BF. The higher the SEE, the lower the precision of the method. Therefore, the obtained SEEs were sorted from smallest to largest and then ranked. The method with the lowest rank (i.e., with the smallest SEE) was considered the most precise.

The ranks obtained from both the (absolute) mean difference to DXA-determined BF and SEE were summed for each anthropometric equation (or set of equations), separately for males and females. The anthropometric equation (or set of equations) with the lowest cumulative rank was selected as the most appropriate for BF assessment.

Finally, the systematic shift from the reference value (DXA), thus bias, was graphically presented with Bland-Altman's scatter plots [34] for each method for body composition assessment (BIA, UWW, and the most appropriate ANT method [see above]), for both males and females.

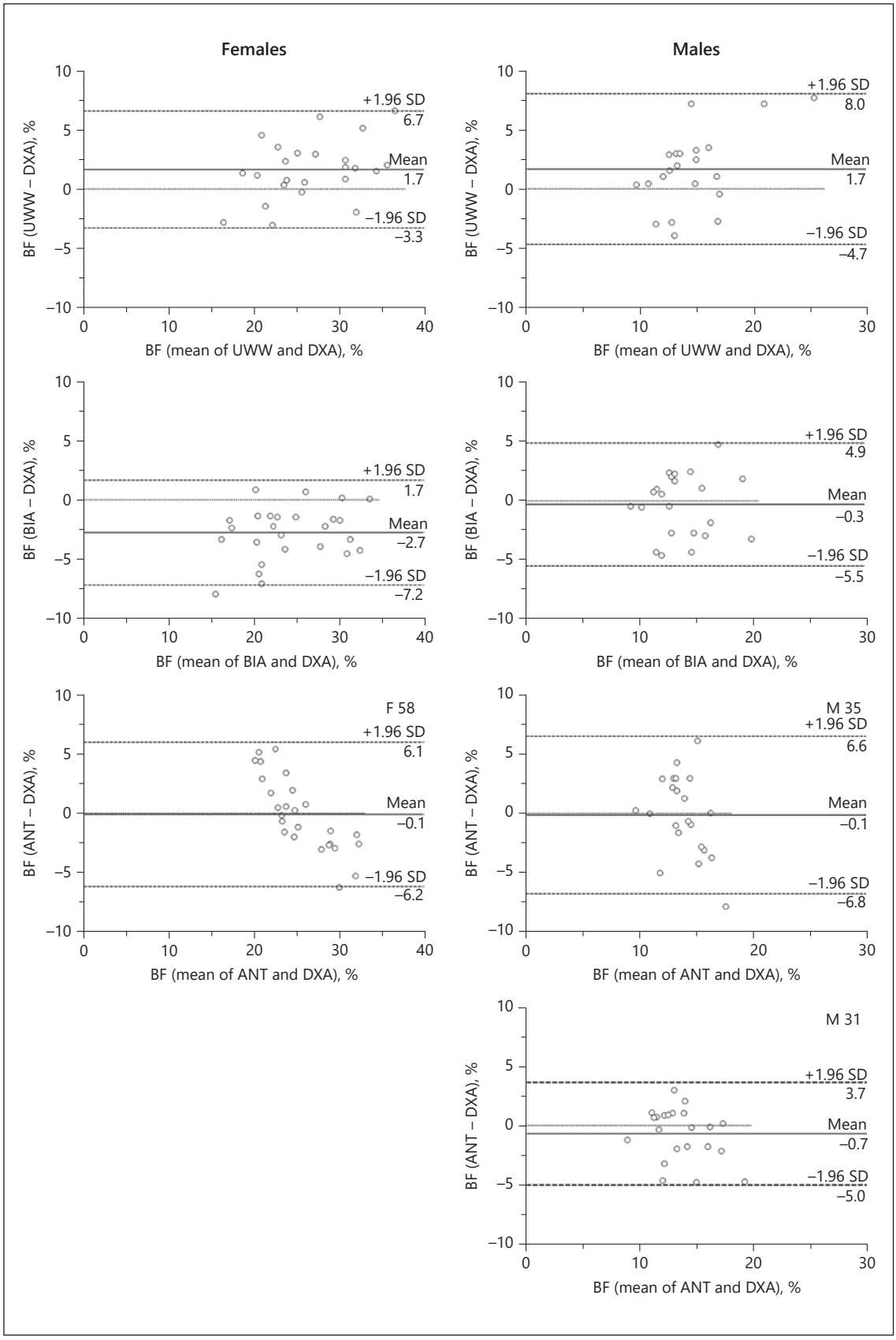
Descriptive statistical analysis was performed with Excel 2010 (Microsoft, USA), computational statistics with MedCalc (12.7.1, Belgium). The level of 0.05 was adopted as statistically significant.

Results

Data are presented as average (SD). Forty-eight subjects participated in the study. Their basic characteristics are presented in Table 1. Skinfold values are presented in Table 2. Subjects' urine specific gravity before BIA measurements was 1.010 (0.007); thus, all subjects were normohydrated.

Fig. 1. Bland-Altman diagrams for body fat percentage (BF; %) obtained with underwater weighing (UWW), bioelectrical impedance measurements (BIA), and anthropometry (ANT), as compared to the reference dual-energy X-ray absorptiometry (DXA), for both females and males. The marks F58, M35, and M31 refer to the anthropometric equations (or sets of equations) used; the exact equations are presented in online supplement 1 and the original equation sources in online supplement 2.

(For figure see next page.)


Females
Males
BF (UWW – DXA), %
BF (mean of UWW and DXA), %
+1.96 SD
6.7
Mean
1.7
–1.96 SD
–3.3
+1.96 SD
8.0
Mean
1.7
–1.96 SD
–4.7
BF (BIA – DXA), %
BF (mean of BIA and DXA), %
+1.96 SD
1.7
Mean
–2.7
–1.96 SD
–7.2
+1.96 SD
4.9
Mean
–0.3
–1.96 SD
–5.5
BF (ANT – DXA), %
BF (mean of ANT and DXA), %
F 58
+1.96 SD
6.1
Mean
–0.1
–1.96 SD
–6.2
M 35
+1.96 SD
6.6
Mean
–0.1
–1.96 SD
–6.8
M 31
+1.96 SD
3.7
Mean
–0.7
–1.96 SD
–5.0

1

The values of BF, as determined with different methods for the assessment of body composition, are presented in Table 3, separately for males and females. For ANT, the results of the regression equation (or set of equations) with the smallest cumulative rank for both, the smallest SEE (precision) and the smallest (absolute) average difference from DXA (accuracy; see Methods), are presented, separately for males and females. According to the above criteria, the set of equations of Lean et al. [25] 1[1] (BD) and Brožek [32][2] (BF) is presented for females. For males, a direct equation of Al-Gindan et al. [30] 2[3] had the lowest cumulative rank for precision and accuracy, and was closely followed by a more precise, but less accurate set of equations of Lean et al. [25] 4[4] (BD) and Brožek [32] (BF).

UWW BF values were higher than the reference DXA BF values in both females ($p = 0.006$) and males ($p = 0.03$). BIA BF values were lower than the reference DXA BF values in females ($p < 0.001$) but did not differ from DXA BF values in males ($p = 0.60$). ANT-determined BF did not differ ($p > 0.05$) from the reference DXA values in both genders, when the methods with the highest combination of accuracy and precision were selected for comparison.

When assessing differences on a group level, Bland-Altman diagrams that compare the 3 methods (UWW, BIA, ANT) with DXA (Fig. 1) demonstrate the largest bias in body fat estimates for UWW for females (an overestimate of 2.9% BF) and the second highest for BIA for females (an underestimate of –2.7% BF). The smallest bias on a group level was observed for the selected anthropometrical methods (0.0 and –0.1% BF for males and females, respectively), when the 2 methods with the highest combination of accuracy and precision were selected; the third smallest bias was observed for BIA in males (–0.3% BF).

BF values, as calculated with a given regression equation (or set of equations), are presented in Figure 2. BF assessed on the same sample of young healthy subjects with different anthropometric equations (or sets of equations) ranged from 8.1 (0.7)% to 28.9 (5.1)% in females and from 5.9 (0.6)% to 28.6 (14.2)% in males.

[1] Lean et al. [25] 1: BD = 1.1145 – (0.000924 × WC) – (0.000465 × AY). BD, body density (g/mL); WC, waist circumference (cm); AY, age (years).
[2] Brožek et al. [32]: BF = 100 × (4.570/BD – 4.142). BF, body fat (%); BD, body density (g/mL).
[3] Al-Gindan et al. [30] 2: (0.158 × BM + 0.383 × WC – 0.118 × BH ×100 – 10.2)/BM) × 100. BM, body mass (kg); WC, waist circumference (cm); BH, body height (m).
[4] Lean et al. [25] 4: BD = 1.1554 – (0.000761 × WC) – (0.00170 × TR) – (0.000532 × AY). BD, body density (g/mL), WC, waist circumference (cm); TR, triceps skinfold (mm); AY, age (years).

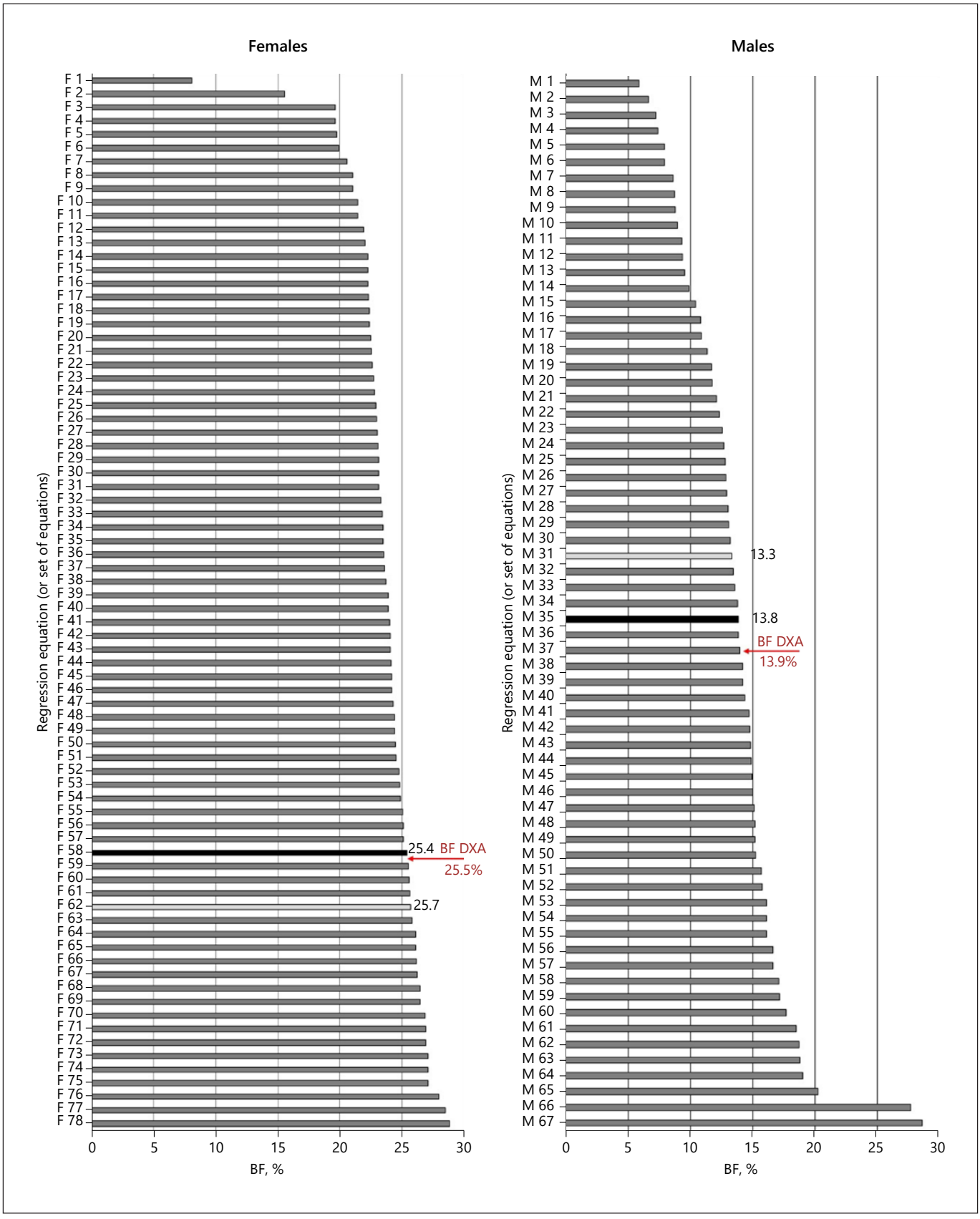
The regression equations (or sets of equations), and rank data for accuracy and precision are presented in online supplement 1. The original published references for the regression equations used are presented in online supplement 2.

Discussion

The present study demonstrated that the results obtained with different methods for the assessment of body composition vary to the extent that the same subject can be (miss)classified into different nutritional categories. When DXA-determined BF is taken as a reference, UWW overestimates BF in young adult healthy subjects of both genders. In contrast, BIA (measured with a given bioimpedance analyser) underestimates BF in females and does not differ from DXA-determined BF in males. In both genders, ANT can provide similar BF estimates than DXA, provided that the appropriate regression equation (or set of equations) is selected. Namely, the results of the study demonstrate a huge difference in BF estimates across a number of the existing anthropometric regression equations (or sets of equations) suitable for the selected sample, provide information about their accuracy and precision, and reveal a list of the most appropriate regression equations (or sets of equations) for use in both young adult healthy females and males. The set of regression equations of Lean et al. [25] 1 and Brožek [32] for females (F58; see online suppl. 1), and both a direct equation of Al-Gindan et al. [30] 2 (M35; see online suppl. 1) and a set of equations of Lean et al. [25] 4 and Brožek [32] (M31; see online suppl. 1) were the methods with the highest combination of accuracy and precision among 145 different equations (or sets of equations) examined (see online suppl. 1, 2). It has to be noted, however, that there were several anthropometric methods for both females and males that did not differ significantly from DXA – their

Fig. 2. Body fat (BF; %) as assessed with different anthropometric equations (or sets of equations); F1–F78 for females, M1–M67 for males. DXA-determined BF value (25.5% for females and 13.9% for males) is marked with an arrow next to the anthropometric equations that yielded the closest BF value. The methods with the highest (black) and second highest (light grey) combination of accuracy and precision relative to DXA-determined BF are presented for both females and males. Asterisks that denote anthropometric methods that differed significantly from DXA are presented in online supplement 1. The original published references for the equations compared are presented in online supplement 2.

(For figure see next page.)


Females
Regression equation (or set of equations)
F 1
F 2
F 3
F 4
F 5
F 6
F 7
F 8
F 9
F 10
F 11
F 12
F 13
F 14
F 15
F 16
F 17
F 18
F 19
F 20
F 21
F 22
F 23
F 24
F 25
F 26
F 27
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F 60
F 61
F 62
F 63
F 64
F 65
F 66
F 67
F 68
F 69
F 70
F 71
F 72
F 73
F 74
F 75
F 76
F 77
F 78
25.4
BF DXA
25.5%
25.7
0
5
10
15
20
25
30
BF, %
Males
Regression equation (or set of equations)
M 1
M 2
M 3
M 4
M 5
M 6
M 7
M 8
M 9
M 10
M 11
M 12
M 13
M 14
M 15
M 16
M 17
M 18
M 19
M 20
M 21
M 22
M 23
M 24
M 25
M 26
M 27
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M 43
M 44
M 45
M 46
M 47
M 48
M 49
M 50
M 51
M 52
M 53
M 54
M 55
M 56
M 57
M 58
M 59
M 60
M 61
M 62
M 63
M 64
M 65
M 66
M 67
13.3
13.8
BF DXA
13.9%
0
5
10
15
20
25
30
BF, %
Color version available online

2

ranks for accuracy and precision are presented in online supplement 1. The selection of an optimal anthropometric method will thus depend on the purpose of each study, taking into account the requirements for its accuracy and precision, as well as the feasibility of anthropometric measurements required for the selected method.

It also seems worth noting that, for females in particular, the Bland-Altman diagram comparing ANT and DXA (Fig. 1) demonstrated a level-dependent bias for the selected anthropometrical method (F58; see online suppl. 2). Namely, although the majority of values on the Bland-Altman diagram lay inside the limits of agreement (±1.96 SD), ANT provided systematically higher body fat values than DXA in females with lower BF, and systematically lower body fat values than DXA in females with higher BF. It therefore seems reasonable to expect that the selected anthropometrical method (F58; see online suppl. 2) may be even further improved.

The literature on body composition assessment provides extensive body composition data for different populations, but the methods used vary considerably. This is a severe drawback, as a direct comparison of data (and thus also of reference values) has been almost impossible to perform. Similarly, although ANT is a widely used method and scientific publications from this field are numerous, the comparison of body composition results obtained with different regression equations has, surprisingly, not been really extensive; most often, only a single or a limited number [21] of selected anthropometrical methods for body composition assessment were compared with a reference method of choice. The purpose of comparisons was most often to check for the accuracy of selected anthropometric method [35, 36], or to create new, population-specific regression equations [37]. Robič et al. [24] performed a comparison of sample-specific and non-specific regression equations for BF calculation in pregnant women and clearly demonstrated that the differences in body composition, as determined with different regression equations (even with those specifically developed for the selected sample), go to the extent that the same subject can easily be (miss)classified into different nutritional categories. This observation has also been proven correct in the present study for a sample of young healthy adults. Namely, for DXA determined BF of 25.5% in females and 13.9% in males, different anthropometric regression equations provided values in a range of 8–29% for females and 6–29% for males. Regression equations for body composition assessment should thus always be applied with sufficient criticism and their selection made according to their accuracy and precision.

A further detailed analysis of individual BF values obtained with different methods might reveal the existence of potential bias patterns for the methods in question; such analysis, however, should be performed on a larger and more variable sample of subjects than in our study, where a rather uniform sample was intentionally used to optimise the comparison of methods and to avoid the necessity of a large sample size due to errors associated with large inter-subject variability. We also believe it would be valuable if the results of our study would in the future be extended with the comparison to multi-compartmental models, such as the one used by Wang et al. [38].

Analysis of potential bias patterns is, however, not achievable for BIA. Namely, different manufacturers of BIA analysers treat the underlying regression equations as business secrets, which is not likely to change, and therefore BIA (as a method on its own) lacks thorough scientific evaluation. To avoid this limitation, some authors have suggested the use of direct BIA measures, thus impedance and resistance values [16, 39]; yet, cut-off impedance and resistance values for underweight, normal weight, overweight, and obese subjects, standardised to a selected reference method, are currently not available. Last, in the present study, UWW could not be reliably performed in some subjects due to their inability to perform a full underwater exhalation, although no non-swimmers were included. This is a factor that will inevitably affect the precision of UWW-determined body composition.

Finally, in the present study DXA was adopted as the reference method. Comparison with DEXA as reference provided information on the relative validity of other methods. DXA certainly has its own limitations, and its results have been shown to be affected by the type of hardware and software used, scanning mode (e.g., pencil beam vs. fan beam), subject thickness [11], differences in calibration, bone edge detection [40], and within a minor extent to hydration status [41]. DXA has also been demonstrated to underestimate BF in comparison to CT, especially in large obese subjects [42]. Yet, DXA has been demonstrated to have the best precision for young healthy subjects [40], as those in the present study. Also, in contrast to impedance and resistance BIA values, at least some gender-specific DXA cut-off values for obesity classification have been reported [43]. DXA also has the advantage of not relying on the assumptions of a 2-compartment model (where body mass is divided into fat mass and fat-free body mass) to provide body composition estimates [11]. Difficulties of accuracy validations for body composition (only a few cadaver studies are available [44,

45]) and continuous advances in DXA technology [46] therefore make DXA a valid choice for body composition assessment. Nevertheless, as our study focused on young healthy adults, and hydration was controlled for, at least some of the above-stated DXA limitations were most likely avoided.

Conclusions

The present study provided grounds for the comparison of body composition results obtained with different methods and across a wide range of anthropometric regression equations. It demonstrated that BF results from different methods for body composition assessment vary to the extent that the same subject can be (miss)classified in different nutritional categories. It revealed that UWW was not only logistically challenging and unfeasible for some subjects, but also overestimated BF compared with DXA. BIA (performed with a given bioimpedance analyser), underestimated BF in women, although it did perform well relative to DXA in males; nevertheless, the method lacks performance standardisation between different conditions and primarily between different BIA devices – an issue that will not change. ANT was proved not to be inferior to DXA, provided that the regression equations with high enough combination of accuracy and precision are chosen; for the time being, this information is only available for young healthy adult subjects. We believe that the extension of the present study to different populations (children, adolescents, elderly adults) would provide a further much needed standardisation for body composition assessment.

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Statement of Ethics

The protocol of the study was approved by the Ethics Committee of the Republic of Slovenia (No. 69/03/15), and the study was performed in accordance with the Declaration of Helsinki. Young healthy volunteers were recruited for participation in the study. They provided informed consent for participation in the study, data collection, data analysis, and presentation of data in written or oral form, for both research and educational purposes, under the condition that their identity is not revealed and that their data are always presented in anonymised form.

Disclosure Statement

S.S. works in a company that is a national sales representative of bioimpedance analysers used in the present study. No financial or any kind of other support for this study was received from the company that produces or sells bioimpedance analysers or any other sources.

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Author Contributions

All authors participated sufficiently in the work to take public responsibility for the content of the paper. P.G. designed and supervised the study, performed data analysis, and wrote the first draft of the manuscript. T.R.P. participated in study design, performed anthropometry measurements, participated in data collection, and reviewed the manuscript. K.Z.K. participated in study design, participated in data collection (BIA), prepared Bland-Altman scatter plots, and reviewed the manuscript. M.F. supervised spirometry measurements and reviewed the manuscript. S.S. participated in study design and reviewed the manuscript. J.K. participated in underwater weighing and reviewed the manuscript. P.K. participated in study design, supervised DXA measurements, and reviewed the manuscript.

All authors have read and approved the final version of the manuscript.

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